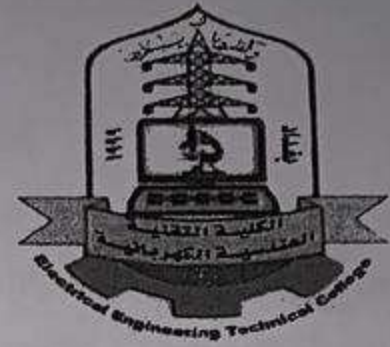




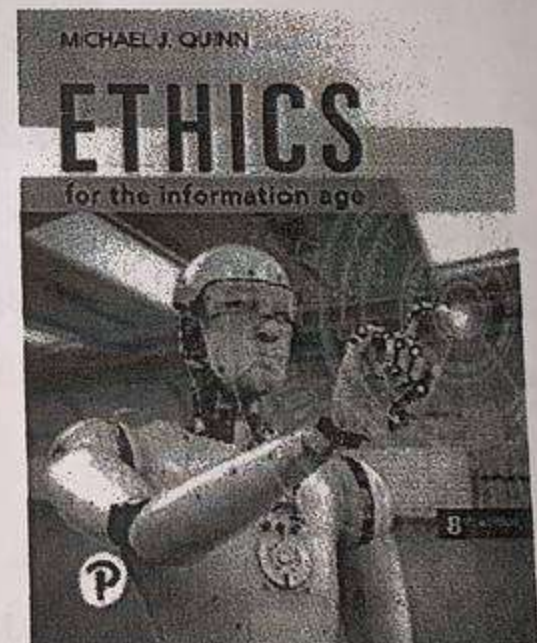
CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

ETHICS FOR THE INFORMATION AGE

FIRST STAGE

Lecture 1

Introduction to computing



Asst. Lect. Haider Saad

2023 - 2024

Introduction to Computing

WE ARE LIVING IN THE INFORMATION AGE. Never before have so many people had such easy access to information. The two principal catalysts for the Information Age have been low-cost computers and high-speed communication networks, which have made possible the development of exciting new technologies, including smartphones, video streaming services, voice-activated digital assistants, low-cost drones, and self-driving cars.

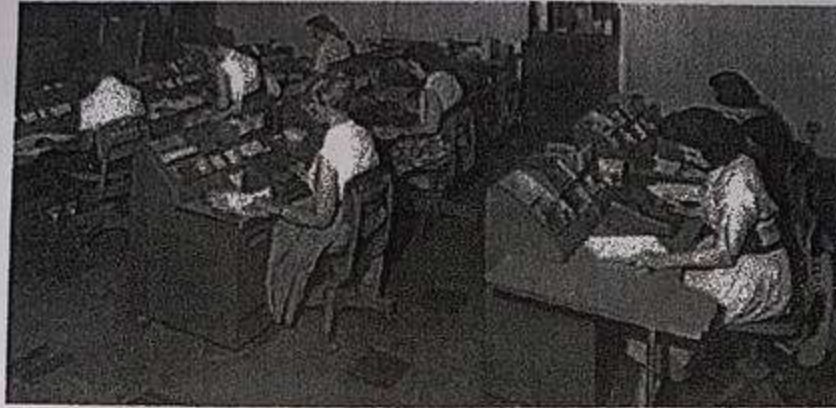


Low-cost computers and high-speed communication networks make possible the products of the Information Age, such as the Samsung Galaxy S9 Plus. It functions as a phone, text messenger, email client, Web browser, camera, video recorder, digital compass, and much more.

1. Mechanical Calculators

- Pascal's calculator, built in 1640, was capable of adding whole numbers containing up to six digits.
- The German Gottfried Leibniz constructed a more sophisticated calculator that could add, subtract, multiply, and divide whole numbers.
- The hand-cranked machine, which he called the Step Reckoner, performed multiplications and divisions through repeated additions and subtractions, respectively.

- The nineteenth century, Leibniz create the Arithmometer, the first commercially successful calculator.
- In contrast, Georg Scheutz and his son Edvard, complete the world's first printing calculator: a machine capable of calculating mathematical tables and typesetting the values onto molds.

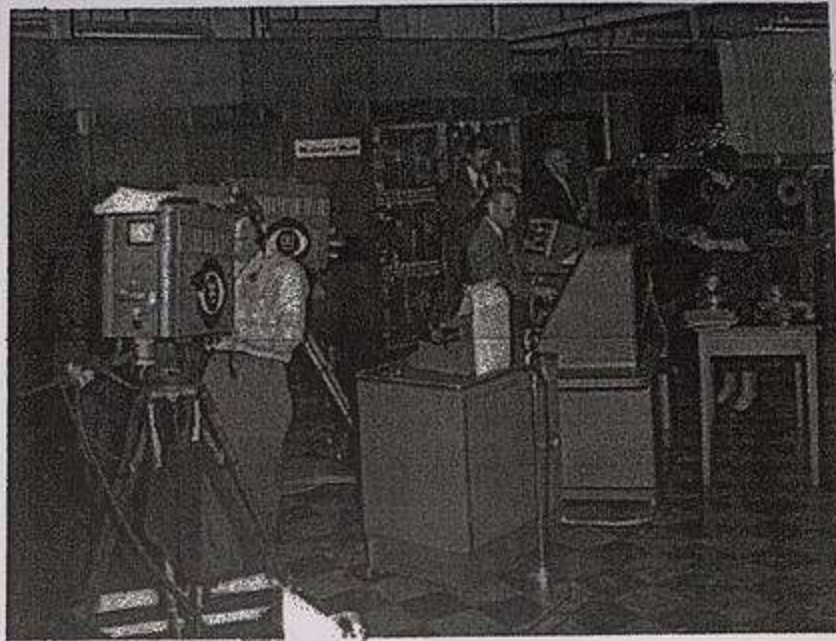


2. Punched-Card Tabulation

- Herman Hollerith created the most successful device. Unlike a predecessor, who chose to record information on rolls of paper, Hollerith decided to record information on punched cards. The use of punched cards to store data was a much better approach because cards could be sorted into groups, allowing the computation of subtotals by categories.
- International Business Machines (IBM) is the corporate descendant of Hollerith's company. Over a period of several decades, IBM and its principal competitor, Remington Rand, developed sophisticated machines based on punched cards: card punches, card verifiers, card tabulators, card sorters, and alphabetizers.
- Customers used these devices to create data-processing systems that received input data, performed one or more calculations, and produced output data.
- Early systems relied on human operators to carry cards from one machine to the next. Later systems had electrical connections that allowed the output of one machine to be transmitted to the next machine without the use of punched cards or human intervention.

3. First Commercial Computers

- In 1951 the British corporation Ferranti Ltd. introduced the Ferranti Mark 1, the world's first commercial computer.
- The Eckert-Mauchly Computer Corporation signed a preliminary agreement with the National Bureau of Standards (representing the Census Bureau) in 1946 to develop a commercial computer, which they called the UNIVAC, for "universal automatic computer".



- In America in the early 1950s, the word "UNIVAC" was synonymous with "computer".

4. Microprocessor

- In 1968 Robert Noyce and Gordon Moore left Fairchild Semiconductor to found another semiconductor manufacturing company, which they named Intel.
- A year later Japanese calculator manufacturer Busicom approached Intel about designing 12 custom chips for use in a new scientific calculator.
- Ted Hoff, Stanley Mazor, and Federico Faggin led to the release of the Intel 4004, the world's first microprocessor. Inside the $1/8$ -inch $\times$ $1/6$ -inch chip were 2,300 transistors, giving the Intel 4004 the same computing power as the ENIAC.

5. Personal Computer

- By the end of the 1970s, many companies, including Apple Computer and Tandy, were producing personal computers.

- In the fall of 1979, Dan Bricklin and Bob Frankston released their program, called VisiCalc, for the Apple II. Sales of the Apple II computer increased significantly after the introduction of VisiCalc.
- The second development was the release of the IBM PC in 1981.
- In 1980 IBM contracted with Microsoft to provide the DOS operating system for the IBM PC.



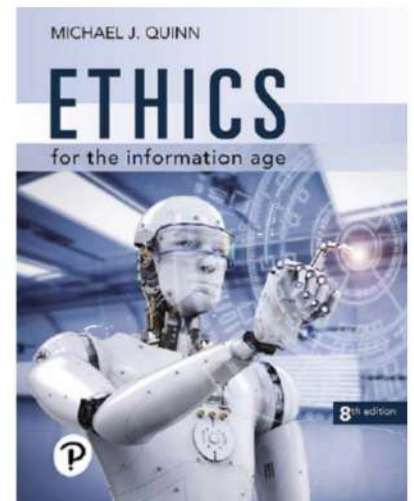
CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

ETHICS FOR THE INFORMATION AGE

FIRST STAGE

Lecture 2

Introduction to Ethics



Asst. Lect. Haider Saad

2023 - 2024

Introduction to Ethics

Defining Terms

1. **Society** is an association of people organized under a system of rules designed to advance the good of its members over time. Every society has rules of conduct describing what people ought and ought not to do in various situations. We call these rules morality.
2. **Ethics** is the philosophical study of morality, a rational examination into people's moral beliefs and behavior. The study of ethics is particularly important right now. Our society is changing rapidly as it incorporates the latest advances in information technology. Ethics is the rational, systematic analysis of conduct that can cause benefit or harm to other people. Because ethics is based in reason, people are required to explain why they hold the opinions they do.

Ethical Theories

1. **Ethical relativism** is the theory that there are no universal moral norms of right and wrong. According to this theory, different individuals or groups of people can have completely opposite views of a moral problem, and both can be right.
Two kinds of ethical relativism: subjective relativism and cultural relativism.
Subjective relativism holds that each person decides right and wrong for himself or herself.
Cultural relativism is the ethical theory that the meaning of "right" and "wrong" rests with a society's actual moral guidelines. These guidelines vary from place to place and from time to time.
2. **Divine Command Theory** The divine command theory is based on the idea that good actions are those aligned with the will of God and bad actions are those contrary to the will of God.
3. **Ethical Egoism** is the philosophy that each person should focus exclusively on his or her self-interest. In other words, according to ethical egoism, the morally right action for a person to take in a particular situation is the action that will provide that person with the maximum long-term benefit.
4. **Kantianism** the name given to the ethical theory of the German philosopher Immanuel Kant. Kant believed that people's actions ought to be guided by moral laws, and that these moral laws were universal. He held that in order to apply to all rational beings, any supreme principle of morality must itself be based on reason.
5. **Act Utilitarianism** the English philosophers Jeremy Bentham (1748–1832) and John Stuart Mill (1806–1873) proposed a theory that is in sharp contrast to Kantianism. According to Bentham and Mill, an action is good if its benefits exceed its harms,

and an action is bad if its harms exceed its benefits. Their ethical theory, called utilitarianism, is based upon the principle of utility, also called the Greatest Happiness Principle.

6. Rule Utilitarianism is the ethical theory that holds that we ought to adopt those moral rules that, if followed by everyone, lead to the greatest increase in total happiness over all affected parties. Hence a rule utilitarian applies the principle of utility to moral rules, while an act utilitarian applies the principle of utility to individual moral actions.
7. Social Contract Theory Morality consists in the set of rules, governing how people are to treat one another that rational people will agree to accept, for their mutual benefit, on the condition that others follow those rules as well.
8. Virtue Ethics a right action is an action that a virtuous person, acting in character, would do in the same circumstances. A virtuous person is a person who possesses and lives out the virtues. The virtues are those character traits human beings need in order to flourish and be truly happy.



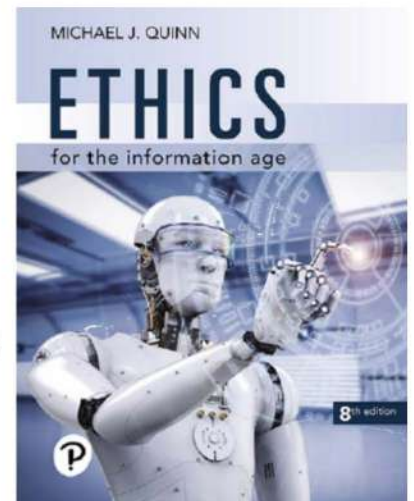
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ETHICS FOR THE INFORMATION AGE

FIRST STAGE

Lecture 3

Networked Communications



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2023 - 2024

Introduction to Ethics

Spam

- About 3.8 billion people around the world now use email, and every day more than 250 billion email messages are sent [6]. Unfortunately, a significant percentage of this traffic consists of unsolicited bulk email, or spam.
- The spam tsunami hit in 2003, with a big jump in the volume of spam messages, and by 2009 about 85 percent of all email messages were spam [10]. Spam consumed a large percentage of the Internet's bandwidth and huge amounts of storage space on mail servers and individual computers.
- To deal with the deluge, an entire industry grew up developing software designed to filter out spam. These filters looked for a large number of messages coming from the same email address, messages with suspicious subject lines, or messages with spamlike content. ISPs and businesses installed spam filters to keep spam from reaching users' mailboxes.
- Where do spammers get email lists with millions of addresses? The Internet provides a variety of sources of email addresses that can be harvested and sold to spammers. For example, email addresses often appear in Web sites, in chat-room conversations, and newsgroups. Some computer viruses gather email addresses stored in the address books of PCs and transmit these addresses to spammers.
- At the peak of the spam tsunami in 2009, about 90 percent of spam was being sent out by botnets: huge networks of compromised computers controlled by "bot herders." Bot herders create botnets by launching programs that search the Internet for computers with inadequate security and install software robot programs, called bots, on these vulnerable systems. A computer with the bot program installed on it is called a zombie because it can be directed by a remote computer to perform certain tasks. Bot herders can send out billions of email messages every day by dividing the address lists among the hundreds of thousands of zombies they control [14].
- Because a spammer's profits increase as the number of sent messages increases, every spammer has an incentive to send as many messages as possible.

Internet Interactions

- The Internet mediates communications and commerce among more than four billion people.

The World Wide Web (WWW)

- The creation of the World Wide Web stimulated a tremendous growth in the popularity of the Internet. Its creator, Tim Berners-Lee, initially proposed the Web as a documentation system for CERN, the Swiss research center for particle physics, but the

creation of easy-to-use Web browsers made the Web accessible to “ordinary” computer users as well [18]. The Web is a hypertext system: a flexible database of information that allows Web pages to be linked to each other in arbitrary fashion. Web browsers such as Chrome, Safari, UC, Firefox, Opera, and Internet Explorer allow people to traverse this hypertext system with ease.

- Two attributes enabled the Web to become a global tool for information exchange. First, it is decentralized. An individual or organization can add new information to the Web without asking for permission from a central authority. Second, every object on the Web has a unique address. Any object can link to any other object by referencing its address. A Web object’s address is called a URL (uniform resource locator).

Mobile Apps

- People are spending more time on smartphones and tablets and less time on laptop or desktop computers. Using Web browsers on mobile devices can be awkward, and for this reason organizations are developing mobile apps: software programs that are loaded onto mobile devices. Some mobile apps are stand-alone programs, but others connect to the Internet, allowing people to download and upload data.
- Mobile apps are becoming an increasingly popular way to access the Internet because they can be optimized to make the best use of a mobile device’s resources (limited screen size, touch interface, etc.).



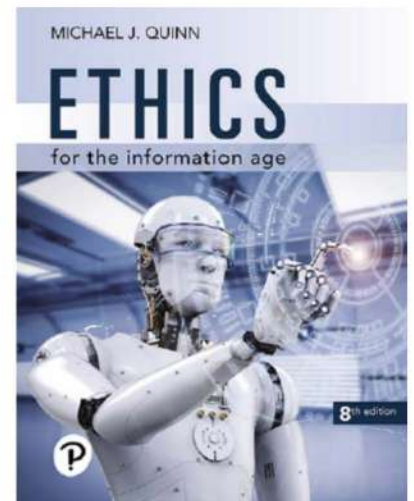
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ETHICS FOR THE INFORMATION AGE

FIRST STAGE

Lecture 4

Impact of Social Media and Online Advertisement



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2023 - 2024

Impact of Social Media and Online Advertisement

Political Activism

- Twitter and Facebook played a highly visible role in the “Arab Spring” demonstrations that led to revolutions in Tunisia and Egypt, a civil war in Libya, and protests in many other Arab countries. In the midst of the protests in Cairo in 2011 that led to the resignation of President Hosni Mubarak, one protester tweeted, “We use Facebook to schedule the protests, Twitter to coordinate, and YouTube to tell the world”. Arab news organization Al Jazeera created a “Twitter Dashboard” indicating the level of tweeting activity in many Arab nations where there was unrest.
- Scholars of the Arab Spring uprisings point to an interesting phenomenon: People started using online social networks such as Twitter in order to keep up with their friends, but these interactions caused them to become politicized. Through these networks, bloggers met new people, became exposed to new ideas, and developed an interest in human rights
- The Arab Spring uprisings demonstrate how Twitter can support grassroots activism. However, Twitter can also be used by a single person or small organization to give the false impression of grassroots support of a political candidate and to trick the Google search engine into prominently featuring misinformation. Here’s an early example of such an effort. In 2010 Massachusetts held a special election to fill its vacant US Senate seat. Someone supporting the Republican candidate created nine fake Twitter accounts and used them to send 979 tweets to 573 users in about two hours. Each of these tweets included the URL of a Web site containing misinformation about the Democratic candidate. These tweets were retweeted 143 times and reached 61,732 Twitter users. Google’s real-time search engine detected the rapidly growing interest in the URL and promoted the URL to the top of its results page for the Democratic candidate. In other words, for a period of time anyone who typed the name of the Democratic candidate into the Google search engine received as the top result the URL of the Web page containing misinformation about that candidate.

Internet Research Agency

- The Internet Research Agency (IRA), based in St. Petersburg, Russia, employs hundreds of people who use social media to exert political influence in other countries. In February 2018, a grand jury for the District of Columbia returned a 37-page indictment charging the IRA, two other Russian organizations, and 13 Russian nationals (ten of them employed by the IRA) with interfering in the US presidential election of 2016, in violation of US law. According to the indictment, the Russians set up social media accounts in the United States using false US personas. Posing as Americans, the Russians posted inflammatory material to social media sites and

organized rallies with the intent of helping the campaign of Donald Trump and harming the campaign of Hillary Clinton.

- The grand jury indictment provided evidence that in addition to the short-term goal of promoting Donald Trump and undermining Hillary Clinton, the IRA's long-term goal was to increase political polarization in the United States. According to the indictment, after Donald Trump was elected president, IRA employees helped organize both pro-Trump and anti-Trump rallies for the same day in New York City.
- Since the election, many have asked whether the "fake news" stories posted by the IRA were responsible for Donald Trump's victory. One study concluded that "fake news most likely did have a substantial impact on the voting decisions of a strategically important set of voters" [53], but another study concluded that social media may have played only a marginal role in influencing voters.

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ETHICS FOR THE INFORMATION AGE

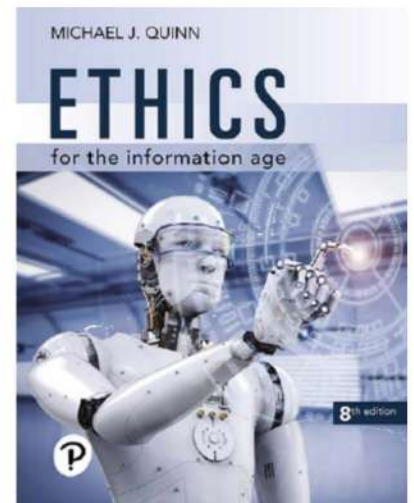
FIRST STAGE

Lecture 4

Children and Inappropriate Contents

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2023 - 2024



Children and Inappropriate Contents

Web Filters

- A *Web filter* is a piece of software that prevents certain Web pages from being displayed by your browser. While you are running your browser, the filter runs as a background process, checking every page your browser attempts to load. If the filter determines that the page is objectionable, it prevents the browser from displaying it.
- Filters can be installed on individual computers, or an ISP may provide filtering services for its customers. Programs designed to be installed on individual computers, such as CyberSentinel and Spector Pro, can be set up to email parents as soon as they detect an inappropriate Web page.
- **Typical filters use two different methods to determine if a page should be blocked.**
The first method is to check the URL of the page against a blacklist of objectionable sites. If the Web page comes from a blacklisted site, it is not displayed.
The second method is to look for combinations of letters or words that may indicate a site has objectionable content.
- Filters can be installed on individual computers, or an ISP may provide filtering services for its customers. Programs designed to be installed on individual computers, such as CyberSentinel and Spector Pro, can be set up to email parents as soon as they detect an inappropriate Web page.
- Neither of these methods is foolproof. The Web contains millions of pages containing pornography, and new sites continue to be created at a high rate, so any blacklist of pornographic sites will be incomplete. Some filters sponsored by conservative groups have blacklisted sites associated with liberal political causes, such as those sponsored by the National Organization for Women and gay and lesbian groups. The algorithms used to identify objectionable words and phrases can cause Web filters to block out legitimate Web pages.

Child Internet Protection Act

- About one-sixth of the libraries in the United States have already installed filtering software on at least some of their computers. The Child Internet Protection Act requires that libraries receiving federal funds to provide Internet access to its patrons must prevent children from getting access to visual depictions of obscenity and child pornography. The law allows adults who desire access to a blocked page to ask a librarian to remove the filter.

Ethical Evaluations of CIPA

Kantian Evaluation

- The goal of CIPA is to protect children from the harm caused by exposure to pornography. The way the goal is being implemented is through Web filters. Studies have demonstrated that Web filters do not block all pornographic material but do block some nonpornographic Web pages.

Act-Utilitarian Evaluation

1. While not all children access the Web in public libraries, and while Web filtering software is imperfect, it is probable that enacting CIPA results in fewer children being exposed to pornography, which is good.
2. Because Web filters are imperfect, people are unable to access some legitimate Web sites. As a result, Web browsers in libraries are less useful as research tools, a harmful consequence.
3. Adult patrons who ask for filters to be removed may be stigmatized (rightly or not) as people who want to view pornography, a harm to them.
4. Some blocked sites may be associated with minority political views, reducing freedom of thought and expression, which is harmful.

Social-Contract-Theory Evaluation

- Analysis from the point of view of social contract theory has produced arguments both supporting and opposing the Children's Internet Protection Act. However, installing filters does not seem to be necessary to preserve the public order. For this reason, the issue is outside the social contract and freedom of conscience should be given precedence.



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ETHICS FOR THE INFORMATION AGE

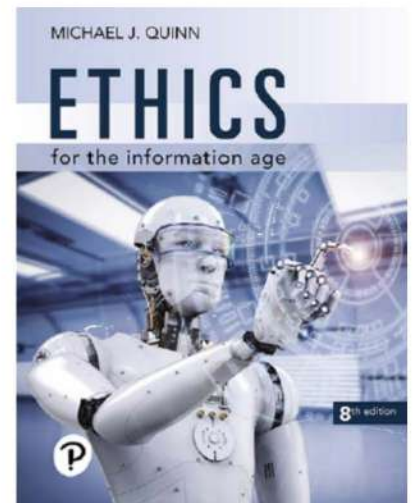
FIRST STAGE

Lecture 6

Intellectual Property

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2023 - 2024



Intellectual Property

Introduction

- Game of Thrones is the most frequently pirated television show in the world. In an effort to reduce piracy, HBO simulcast the premiere episode of season 5 in 170 countries and introduced a new video-streaming service for \$14.99 a month with a 30-day free trial. A record 8 million viewers watched the first episode on HBO. However, a day before it aired, the first four episodes became available on pirate networks, and in a single week were downloaded more than 32 million times worldwide. Is it fair for some people to watch shows for free when others are paying to view them?
- As a society we benefit from access to high-quality television shows, music, movies, computer programs, and other products of the human intellect. The value of these intellectual properties is much higher than the cost of the media on which they are distributed, tempting people to make unauthorized copies. When this happens, producers of intellectual property do not receive all the payments the law says they are entitled to.

Property Rights

- The English philosopher John Locke (1632–1704) developed an influential theory of property rights. In *The Second Treatise of Government*, Locke makes the following case for a natural right to property:
First: people have a right to property in their own person. Nobody has a right to the person of anybody else.
Second: people have a right to their own labor. The work that people perform should be to their own benefit.
Third: people have a right to those things that they have removed from Nature through their own labor.
- To Locke, the definition of ownership is fulfilled as long as these two conditions are met:
 1. No person claims more property than he or she can use.
 2. When people remove something from the common state in order to make it their own property, there is still plenty left over for others to claim through their labor

Benefits of Intellectual Property Protection

- The US Constitution gives Congress the power to promote the Progress of Science and useful Arts by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.
- If a person has the right to control the distribution and use of a piece of intellectual property, there are many opportunities for that person to make money.

Limits to Intellectual Property Protection

- Society benefits the most when inventions are in the public domain and anyone can take advantage of them.
- Granted authors and inventors exclusive rights to their writings and discoveries but only for a limited period of time and at the end of that time period, the intellectual property enters the public domain

Asst. Lect. Haider Saad



CYBER SECURITY TECHNOLOGY ENGINEERING DEPARTMENT

ETHICS FOR THE INFORMATION AGE

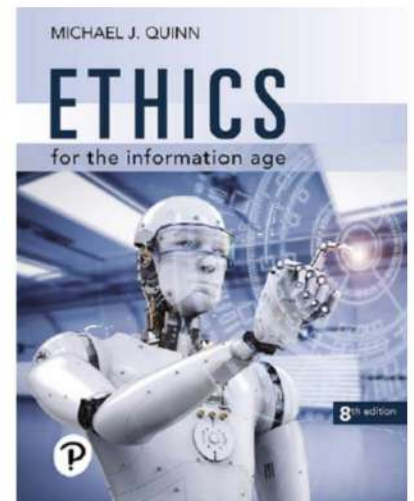
FIRST STAGE

Lecture 7

Trademark, Patents, and Copyright

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2023 - 2024



Trademark, Patents, and Copyright

Trademark

- A **trademark** is a word, symbol, picture, sound, or color used by a business to identify goods. A service mark is a mark identifying a service. Hereafter, we will use the word “trademark” to mean either a trademark or a service mark. By granting a trademark, a government gives a company the right to use it and the right to prevent other companies from using it. Through the use of a trademark, a company can establish a “brand name.” Society benefits from branding because branding allows consumers to have more confidence in the quality of the products and services they purchase.
- Obtaining a trademark requires choosing the mark, identifying the goods and/or services that will be identified by the mark, searching the US Patent and Trademark Office database to determine if a similar mark has already been registered for comparable goods and/or services, and demonstrating that the mark is being used or about to be used in commerce.

Patent

- A patent is how the US government provides intellectual property protection for a limited period of time to creators of machines, systems, and other inventions. A patent is quite different from a trade secret because a patent is a public document that provides a detailed description of the invention. The owner of the patent can prevent others from making, using, or selling the invention for the lifetime of the patent, after the patent expires, anyone has the right to make use of its ideas.

Copyrights

- A copyright is how the US government provides authors with certain rights to original works that they have written. The owner of a copyright has five principal rights:
 1. The right to reproduce the copyrighted work.
 2. The right to distribute copies of the work to the public.
 3. The right to display copies of the work in public.
 4. The right to perform the work in public.
 5. The right to produce new works derived from the copyrighted work.



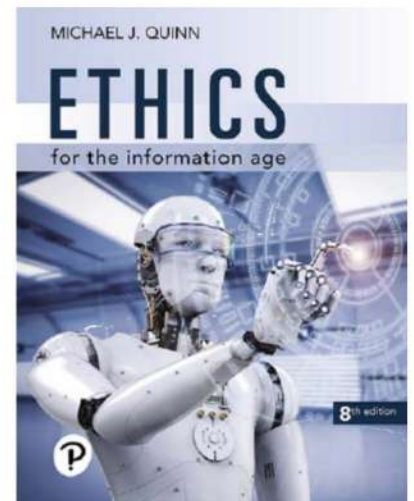
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ETHICS FOR THE INFORMATION AGE

FIRST STAGE

Lecture 8, 9

Protections for Software & Open Source Software



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2023 - 2024

Protections for Software

Software Copyrights

- The first software copyrights were applied for in 1964.
- When a piece of software gets copyright protection, what exactly is copyrighted?
First: copyright protects the original expression of an idea, not the idea itself.
Second: copyright usually protects the executable program, not the source program. Typically, the source code to a program is confidential.
- Doing any of the following actions without the authorization of the copyright holder is a violation of copyright law:
 1. Copying a program onto a CD to give or sell to someone else.
 2. Preloading a program onto the hard disk of a computer being sold.
 3. Distributing a program over the Internet.

Open Source Software

- **Open source:** is an alternative way of distributing software.
- **Benefits of Open Source Software:**
 1. It gives everyone using a program the opportunity to improve it.
 2. Users of open-source programs do not have to wait as long for bug fixes and patches.
 3. It eliminates the tension between obeying copyright law and helping others.
 4. open-source programs are the property of the entire user community, not just a single vendor.
 5. It shifts the focus from manufacturing to service, which can result in customers getting better support for their software.
- **Examples of Open-Source Software:**
 1. The Android operating system is the world's best-selling smartphone platform.
 2. Chrome is the world's most popular Web browser, and Firefox is the number-four browser, according to StatCounter.
 3. Other popular open-source programming languages and tools are Python, Ruby, TCL/TK and PHP.
- **Impact of Open-Source Software:**

The production of Linux operating system was by several thousand developers and hackers scattered all over the planet, connected only by the internet.

Q1. Fill in the following blanks with appropriate ones:

1. ... **Low-cost computers** ... and ... **high-speed communication networks** ... networks make possible the products of the Information Age, such as the Samsung Galaxy S9 Plus.
2. ... **Pascal's calculator** ... built in 1640, was capable of adding whole numbers containing up to six digits.
3. Unlike a predecessor, who chose to record information on rolls of paper, Hollerith decided to record information on ... **punched cards**....
4. ... **Cultural relativism**.. is the ethical theory that the meaning of “right” and “wrong” rests with a society’s actual moral guidelines.
5. ... **Virtue Ethics** a right action is an action that a virtuous person, acting in character, would do in the same circumstances.
6. ISPs and businesses installed ... **spam filters**... to keep spam from reaching users’ mailboxes.
7. The hand-cranked machine, which he called ... **the Step Reckoner**... performed multiplications and divisions through repeated additions and subtractions, respectively.
8. The nineteenth century, Leibniz create the ... **Arithmometer** ... the first commercially successful calculator.
9. UNIVAC stands for ... **universal automatic computer**
10. In 1968 Robert Noyce and Gordon Moore left Fairchild Semiconductor to found another semiconductor manufacturing company, which they named ... **Intel** ...
11. By the end of the 1970s, many companies, including ... **Apple Computer**... and ...**Tandy**... were producing personal computers.
12. In the fall of 1979, Dan Bricklin and Bob Frankston released their program, called ...**VisiCalc**... for the Apple II.
13. In 1980 IBM contracted with Microsoft to provide the ...**DOS**... operating system for the IBM PC.
14. Two kinds of ethical relativism: ...**subjective relativism**... and ...**cultural relativism**...
15. Every society has rules of conduct describing what people ought and ought not to do in various situations. We call these rules ...**morality**...
16. Utilitarianism, is based upon the principle of utility, also called ...**the Greatest Happiness Principle**...
17. A computer with the bot program installed on it is called a ...**zombie**...
18. URL stands for ...**uniform resource locator**...
19. ...**Spam**... consumed a large percentage of the Internet’s bandwidth and huge amounts of storage space on mail servers and individual computers.
20. WWW stands for ...**World Wide Web**...
21. ...**Web filter**... is a piece of software that prevents certain Web pages from being displayed by your browser.
22. Through the use of a trademark, a company can establish a ...**brand name**...
23. Patent is quite different from a ...**trade secret**... because a patent is a public document that provides a detailed description of the invention.

Q2. Define the Ethics and list its theories.

Ethics is the philosophical study of morality, a rational examination into people's moral beliefs and behavior. The study of ethics is particularly important right now. Our society is changing rapidly as it incorporates the latest advances in information technology.

Or **Ethics** is the rational, systematic analysis of conduct that can cause benefit or harm to other people. Because ethics is based in reason, people are required to explain why they hold the opinions they do.

Ethical Theories:

1. Ethical relativism.
2. Divine Command Theory.
3. Ethical Egoism.
4. Kantianism.
5. Act Utilitarianism.
6. Rule Utilitarianism.
7. Social Contract Theory.
8. Virtue Ethics.

Q3. Define each of the following:

- | | | | |
|------------------------|---------------|-----------------------|-------------------|
| 1. Society | 2. Ethics | 3. Ethical relativism | 4. Ethical Egoism |
| 5. Rule Utilitarianism | 6. botnets | 7. Web filter | 8. Trademark |
| 9. Patent | 10. Copyright | 11. Open source | |

1. **Society** is an association of people organized under a system of rules designed to advance the good of its members over time. Every society has rules of conduct describing what people ought and ought not to do in various situations. We call these rules morality.

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6. **Botnets** huge networks of compromised computers controlled by "bot herders".

7. **Web filter** is a piece of software that prevents certain Web pages from being displayed by your browser. While you are running your browser, the filter runs as a background process, checking every page your browser attempts to load. If the filter determines that the page is objectionable, it prevents the browser from displaying it.

8. **Trademark** is a word, symbol, picture, sound, or color used by a business to identify goods.

9. **Patent** is how the US government provides intellectual property protection for a limited period of time to creators of machines, systems, and other inventions.

10. **Copyright** is how the US government provides authors with certain rights to original works that they have written.

11. **Open source** is an alternative way of distributing software.

Q4. What is two kinds of ethical relativism?

Subjective relativism holds that each person decides right and wrong for himself or herself.

Cultural relativism is the ethical theory that the meaning of “right” and “wrong” rests with a society’s actual moral guidelines. These guidelines vary from place to place and from time to time.

Q5. What is the two attributes enabled the Web to become a global tool for information exchange?

First, it is decentralized. An individual or organization can add new information to the Web without asking for permission from a central authority.

Second, every object on the Web has a unique address. Any object can link to any other object by referencing its address. A Web object’s address is called a URL (uniform resource locator).

Q6. Where do spammers get email lists with millions of addresses?

The Internet provides a variety of sources of email addresses that can be harvested and sold to spammers. For example, email addresses often appear in Web sites, in chat-room conversations, and newsgroups. Some computer viruses gather email addresses stored in the address books of PCs and transmit these addresses to spammers.

Q7. What is the two different methods used by typical filters to determine if a page should be blocked?

The first method is to check the URL of the page against a blacklist of objectionable sites. If the Web page comes from a blacklisted site, it is not displayed.

The second method is to look for combinations of letters or words that may indicate a site has objectionable content.

Q8. What is the Ethical Evaluations of CIPA according to Act-Utilitarian Evaluation?

1. While not all children access the Web in public libraries, and while Web filtering software is imperfect, it is probable that enacting CIPA results in fewer children being exposed to pornography, which is good.

2. Because Web filters are imperfect, people are unable to access some legitimate Web sites. As a result, Web browsers in libraries are less useful as research tools, a harmful consequence.

3. Adult patrons who ask for filters to be removed may be stigmatized (rightly or not) as people who want to view pornography, a harm to them.

4. Some blocked sites may be associated with minority political views, reducing freedom of thought and expression, which is harmful.

Q9. What is the Ethical Evaluations of CIPA according to Kantian?

The goal of CIPA is to protect children from the harm caused by exposure to pornography. The way the goal is being implemented is through Web filters. Studies have demonstrated that Web filters do not block all pornographic material but do block some nonpornographic Web pages.

Q10. What is the Ethical Evaluations of CIPA according to Social-Contract-Theory?

Analysis from the point of view of social contract theory has produced arguments both supporting and opposing the Children's Internet Protection Act. However, installing filters does not seem to be necessary to preserve the public order. For this reason, the issue is outside the social contract and freedom of conscience should be given precedence.

Q11. What is the case for a natural right to property that Locke makes?

First: people have a right to property in their own person. Nobody has a right to the person of anybody else.

Second: people have a right to their own labor. The work that people perform should be to their own benefit.

Third: people have a right to those things that they have removed from Nature through their own labor.

Q12. What are the two conditions that fulfilled the definition the ownership according to Locke?

1. No person claims more property than he or she can use.
2. When people remove something from the common state in order to make it their own property, there is still plenty left over for others to claim through their labor.

Q13. What are the benefits of Intellectual Property Protection?

- The US Constitution gives Congress the power to promote the Progress of Science and useful Arts by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.

-If a person has the right to control the distribution and use of a piece of intellectual property, there are many opportunities for that person to make money.

Q14. What are the limits of Intellectual Property Protection?

-Society benefits the most when inventions are in the public domain and anyone can take advantage of them.

-Granted authors and inventors exclusive rights to their writings and discoveries but only for a limited period of time and at the end of that time period, the intellectual property enters the public domain.

Q15. What does it require to obtain a trademark?

1. Choosing the mark.
2. Identifying the goods and/or services that will be identified by the mark.
3. Searching the US Patent and Trademark Office database to determine if a similar mark has already been registered for comparable goods and/or services.

Q16. What are the five main rights that a copyright owner has?

1. The right to reproduce the copyrighted work.
2. The right to distribute copies of the work to the public.
3. The right to display copies of the work in public.
4. The right to perform the work in public.
5. The right to produce new works derived from the copyrighted work.

Q17. When a piece of software gets copyright protection, what exactly is copyrighted?

First: copyright protects the original expression of an idea, not the idea itself.

Second: copyright usually protects the executable program, not the source program. Typically, the source code to a program is confidential.

Q18. What are the procedures that constitute a violation of copyright law?

1. Copying a program onto a CD to give or sell to someone else.
2. Preloading a program onto the hard disk of a computer being sold.
3. Distributing a program over the Internet.

Q19. What are the Benefits of Open Source Software?

1. It gives everyone using a program the opportunity to improve it.
2. Users of open-source programs do not have to wait as long for bug fixes and patches.
3. It eliminates the tension between obeying copyright law and helping others.
4. open-source programs are the property of the entire user community, not just a single vendor.
5. It shifts the focus from manufacturing to service, which can result in customers getting better support for their software.

Q20. Choose the correct answer:

1. Played a highly visible role in the “Arab Spring” demonstrations.
a. Facebook b. Twitter c. YouTube d. All of the previous
2. Employs hundreds of people who use social media to exert political influence in other countries.
a. ISP b. URL c. IRA d. All of the previous
3. a piece of software that prevents certain Web pages from being displayed by your browser.
a. Spam filters b. Web filters c. System filters d. Email filters
4. Ethical evaluation of CIPA which stipulates: The goal of CIPA is to protect children from the harm caused by exposure to pornography.
a. Kantian b. Act-Utilitarian c. Social-Contract-Theory d. All of the previous
5. Botnets: huge networks of compromised computers controlled by:
a. bot herders b. Robot net c. robot programs d. None of the previous
6. The Web is:
a. Internet system b. Hypertext system c. Operating system d. None of the previous
7. Is the rational, systematic analysis of conduct that can cause benefit or harm to other people
a. Society b. Morality c. Ethics d. None of the previous
8. The word “UNIVAC” was synonymous with:
a. Microprocessor b. First calculator c. Punched machine d. Computer
9. Popular open-source programming languages and tools are
a. Android b. Google c. Firefox d. Python
10. Is the world’s most popular Web browser
a. Opera b. Internet explorer c. Chrome d. Firefox